

Lesson 1. Set Theory, Logic and Induction

FU03 – Discrete Systems

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June 12, 2026

- **Lectures:** VIGLIETTA Giovanni (vigliett@u-aizu.ac.jp)
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- **Exercises:** NASSANI Alaeddin (alaeddin@u-aizu.ac.jp)
Tuesdays and Fridays, Period 8, Lecture Theater (both C5 and C6)
- **Grading method:** Assignments ($3 \times 10\%$) + Final exam (70%)
- Lecture slides, announcements, etc. are available on LMS:
<https://elms.u-aizu.ac.jp/course/view.php?id=7721>
- This course is offered in English.



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Course topics

- Elementary set theory (Chapter 1)
- Propositional logic (Chapter 4)
- Mathematical induction (Chapter 1)
- Functions (Chapter 3)
- Relations (Chapter 2)
- Counting (Chapter 5)
- Recurrence relations (Chapter 6)
- Integers and divisibility (Chapter 11)
- Modular arithmetic (Chapter 11)
- Graph theory (Chapter 8)
- Directed graphs (Chapter 9)
- Ordered sets (Chapter 14)
- Boolean algebra (Chapter 15)

Today's topics

- Elementary set theory (Chapter 1)
- Propositional logic (Chapter 4)
- Mathematical induction (Chapter 1)

Set definition and notation

In elementary set theory, a *set* is simply a collection of objects called *elements*.

- $a \in S$: the object a is an element of the set S
- $b \notin S$: the object b is not an element of the set S

A set can be defined in two ways:

- By listing its elements: $S = \{1, 3, 5, 7, 9\}$
- By a property: $S = \{x \mid x \text{ is an odd integer, } 0 \leq x \leq 10\}$

Note. All of these sets are *the same set*:

- $\{1, 2, 3\}$
- $\{2, 3, 1\}$
- $\{1, 2, 2, 3\}$
- $\{3, 2, 1, 3, 2, 1, 3, 2, 1\}$
- $\{1, (1 + 1), (1 + 1 + 1)\}$

Finite and infinite sets

A set is *finite* if it has exactly n elements, where n is an integer. The number of elements of S is denoted as $\#S$ or $|S|$.

- $\#\{1, 2, 3, 4, 5\} = 5$
- $\#\{2, 2, 2, 2, 1, 4\} = 3$
- $\#\{x \mid x \text{ is an integer, } 1 \leq x \leq 10\} = 10$

A special finite set is the *empty set*: $\emptyset = \{\}$. Note that $\#\emptyset = 0$.

A set that is not finite is called *infinite*.

- Natural numbers: $\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$
- Integer numbers: $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- Rational numbers (fractions): $\mathbb{Q} = \{\frac{1}{3}, \frac{5}{2}, -\frac{3}{4}, 0, -\frac{384}{65}, 5, \dots\}$
- Real numbers: $\mathbb{R} = \{0, 1, -2, \frac{1}{2}, -\frac{1}{3}, \sqrt{2}, \pi, e, \log_2 10, \dots\}$

If S is an infinite set, we write $\#S = \infty$.

If all elements of A are also elements of B , then A is a *subset* of B . In symbols, $A \subseteq B$.

In other words, $A \subseteq B$ if and only if $a \in A$ implies $a \in B$.

Example. $\emptyset \subseteq \{1\} \subseteq \{1, 2\} \subseteq \mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$

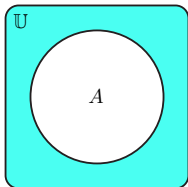
Note. $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Note. If $A \subseteq B$, then $|A| \leq |B|$.

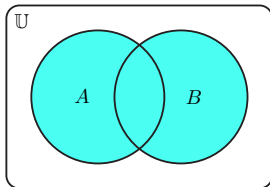
Operations on sets

We define the following well-known operations on sets:

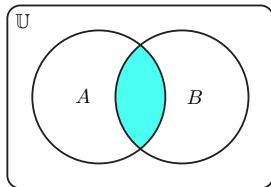
Complement: $\bar{A}$



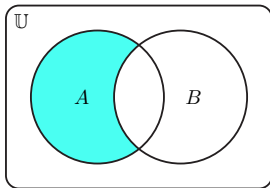
Union: $A \cup B$



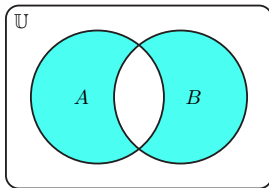
Intersection: $A \cap B$



Difference: $A \setminus B$



Symmetric difference: $A \oplus B$



We denote as $\mathbb{U}$ the *universal set*, which contains all elements.

Exercise

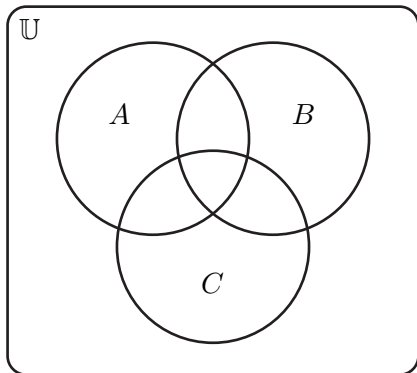
Identify the following sets in the diagram:

- $(A \cup B) \cap C$

- $A \cap (B \setminus C)$

- $\overline{(A \setminus B)} \cup C$

- $(A \oplus B) \oplus \bar{C}$



Principle of duality

Some notable properties of set operations:

Idempotence	$A \cup A = A$	$A \cap A = A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Commutativity	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Distributivity	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \emptyset = A$ $A \cup \mathbb{U} = \mathbb{U}$	$A \cap \mathbb{U} = A$ $A \cap \emptyset = \emptyset$
Involution	$\overline{\overline{A}} = A$	
Complement	$A \cup \overline{A} = \mathbb{U}$ $\overline{\overline{U}} = \emptyset$	$A \cap \overline{A} = \emptyset$ $\overline{\emptyset} = \mathbb{U}$
DeMorgan	$\overline{A \cup B} = \overline{A} \cap \overline{B}$	$\overline{A \cap B} = \overline{A} \cup \overline{B}$

In general, from a true equation involving the above symbols, we can obtain another true equation by swapping $\cup \leftrightarrow \cap$ and $\emptyset \leftrightarrow \mathbb{U}$

Partitions, set product, and power set

- A *partition* of a set S is a subdivision of S into non-empty disjoint subsets:

$$S = A_1 \cup A_2 \cup A_3 \cup \dots, \text{ and } A_i \neq A_j \text{ implies } A_i \cap A_j = \emptyset$$

- If A and B are two sets, their *product* $A \times B$ is the set of all pairs (a, b) with $a \in A$ and $b \in B$.

Example. If $A = \{a, b, c\}$ and $B = \{a, d\}$, then

$$A \times B = \{(a, a), (a, d), (b, a), (b, d), (c, a), (c, d)\}$$

Note. $|A \times B| = |A| \cdot |B|$

- The *power set* $\mathcal{P}(S)$ is the set of all subsets of S .

Example. If $S = \{1, 2, 3\}$, then

$$\mathcal{P}(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, S\}$$

Note. If $|S| = n$, then $|\mathcal{P}(S)| = 2^n$.

Propositions and predicates

- A *proposition* is a statement that is either true or false.

Example. $p \equiv$ There are 7 days in a week

Example. $q \equiv$ London is the capital of France

- A *predicate* is a statement with some free variables. Assigning values to the variables turns a predicate into a proposition.

Example. $p(x) \equiv x > 3$

Then, $p(5) \equiv 5 > 3$ (true), and $p(2) \equiv 2 > 3$ (false)

Example. $q(x, y) \equiv x$ is an integer multiple of y

Then, $q(15, 3)$ is true, and $q(9, 5)$ is false

Logical operators

We can connect propositions and predicates by operators:

- **Negation:** $\neg p$ (read: “not p ”)
- **Conjunction:** $p \wedge q$ (read: “ p and q ”)
- **Disjunction:** $p \vee q$ (read: “ p or q ”)

p	$\neg p$
T	F
F	T

p	q	$p \wedge q$	$p \vee q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

How do we evaluate a compound proposition or predicate?
By writing its *truth table*.

Exercise

Write the truth table of the compound proposition $\neg(p \wedge \neg q) \vee r$.

p	q	r	$\neg q$	$p \wedge \neg q$	$\neg(p \wedge \neg q)$	$\neg(p \wedge \neg q) \vee r$
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

Logic vs. set theory

If two (compound) propositions p and q have the same truth table, then p and q are *equivalent*, and we write $p \equiv q$.

Example. $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

Recall the distributive law for sets:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Notice any similarity? This is not a coincidence...

- We can turn any predicate $p(x)$ into a set: $\mathcal{T}_p = \{x \mid p(x)\}$
- We can turn any set S into a predicate: $\varphi_S(x) \equiv x \in S$

So, we can turn predicates into sets and vice versa.
But there is more...

Logic vs. set theory

There is a correspondence between set-theoretic operations and logical operators:

- $\mathcal{T}_{\neg p} = \overline{\mathcal{T}_p}$
- $\neg \varphi_A(x) \equiv \varphi_{\bar{A}}(x)$
- $\mathcal{T}_{p \vee q} = \mathcal{T}_p \cup \mathcal{T}_q$
- $\varphi_A(x) \vee \varphi_B(x) \equiv \varphi_{A \cup B}(x)$
- $\mathcal{T}_{p \wedge q} = \mathcal{T}_p \cap \mathcal{T}_q$
- $\varphi_A(x) \wedge \varphi_B(x) \equiv \varphi_{A \cap B}(x)$

So, we can turn any set equation into a proposition equivalence, and vice versa. In particular, for propositional logic there is a *principle of duality* similar to that of set theory:

From any equivalence involving $\neg$, $\wedge$, $\vee$, $\top$, $\bot$, we can obtain another equivalence by swapping $\wedge \longleftrightarrow \vee$ and $\top \longleftrightarrow \bot$

Example. $p \vee \bot \equiv p$, and so $p \wedge \top \equiv p$

Tautologies and contradictions

Some propositions are necessarily true or necessarily false:

- If $p \equiv \text{T}$, then p is a *tautology*.
- If $p \equiv \text{F}$, then p is a *contradiction*.

Example. $p \vee \neg p$ is a tautology; $p \wedge \neg p$ is a contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

The dual of a tautology is always a contradiction, and vice versa.

Implication and coimplication

There are two more logical operators:

- **Implication:** $p \rightarrow q$ (read: “ p implies q ” or “if p then q ”)
- **Coimplication:** $p \leftrightarrow q$ (read: “ p if and only if q ”)

p	q	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	T

Note the following properties:

- $p \rightarrow q \equiv \neg p \vee q$
- $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
- $p \leftrightarrow q$ if and only if $p \equiv q$

Logical quantifiers

We can attach *quantifiers* to predicates:

- **Existential:** $\exists x$ (read: “there exists x ”)
- **Universal:** $\forall x$ (read: “for every x ”)

Example. $\exists x \in \mathbb{N}, x > 10$

This is a true proposition: $x = 11$ is an example.

Example. $\forall x \in \mathbb{N}, x > 10$

This is a false proposition: $x = 9$ is a counterexample.

There is a relationship between quantifiers and sets:

- $\exists x, p(x) \equiv \mathcal{T}_p \neq \emptyset$
- $\forall x, p(x) \rightarrow q(x) \equiv \mathcal{T}_p \subseteq \mathcal{T}_q$
- $\forall x, p(x) \equiv \mathcal{T}_p = \mathbb{U}$
- $\forall x, p(x) \leftrightarrow q(x) \equiv \mathcal{T}_p = \mathcal{T}_q$

There are also DeMorgan's laws for quantifiers:

- $\neg \exists x, p(x) \equiv \forall x, \neg p(x)$
- $\neg \forall x, p(x) \equiv \exists x, \neg p(x)$

Mathematical induction

Let's add up the first few powers of 2.

$$2^0 = 1 = 1$$

$$2^0 + 2^1 = 1 + 2 = 3$$

$$2^0 + 2^1 + 2^2 = 1 + 2 + 4 = 7$$

$$2^0 + 2^1 + 2^2 + 2^3 = 1 + 2 + 4 + 8 = 15$$

$$2^0 + 2^1 + 2^2 + 2^3 + 2^4 = 1 + 2 + 4 + 8 + 16 = 31$$

$$2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 = 1 + 2 + 4 + 8 + 16 + 32 = 63$$

Do you notice a pattern?

Mathematical induction

Perhaps $2^0 + 2^1 + 2^2 + \dots + 2^{n-1} = 2^n - 1$ for all $n \in \mathbb{N}$?

How can we prove that? Certainly we can't enumerate all cases...

We already know that this equation is true for $n \in \{1, 2, 3, 4, 5, 6\}$.

Can we extend it in some general way?

The last equation we found was

$$2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 = 2^6 - 1$$

Now we want to compute

$$2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6$$

Can we re-use what we already know?

$$(2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5) + (2^6) = (2^6 - 1) + (2^6) = 2 \cdot 2^6 - 1 = 2^7 - 1$$

So the equation is true for $n = 7$, too!

Did we find a general method?

Mathematical induction

We want to prove $2^0 + 2^1 + 2^2 + \cdots + 2^{n-1} = 2^n - 1$ for all $n \in \mathbb{N}$.

Assume we know that the equation holds for some generic $n = k$.

$$2^0 + 2^1 + 2^2 + \cdots + 2^{k-1} = 2^k - 1$$

Can we prove that it also holds for $n = k + 1$?

$$2^0 + 2^1 + 2^2 + \cdots + 2^{k-1} + 2^k = (2^k - 1) + (2^k) = 2^{k+1} - 1$$

We knew that the equation was true for $n = k$,
and we used this knowledge to prove it for $n = k + 1$.

Since k is a generic number, this means we can also prove it for $n = k + 2$, and for $n = k + 3$, and for $n = k + 4$, and so on.

We conclude that the equation holds for all $n \in \mathbb{N}$.

This proof technique is called *mathematical induction*.

Mathematical induction

In general, mathematical induction works like this:

We want to prove some predicate $P(n)$ for all $n \in \mathbb{N}$.

- We prove $P(1)$ directly
- We assume that $P(k)$ is true, and we prove $P(k + 1)$

This makes $P(n)$ automatically true for all $n \in \mathbb{N}$.

Indeed, $P(1)$ implies $P(2)$, which implies $P(3)$, which implies $P(4)$, which implies $P(5)$, and so on.

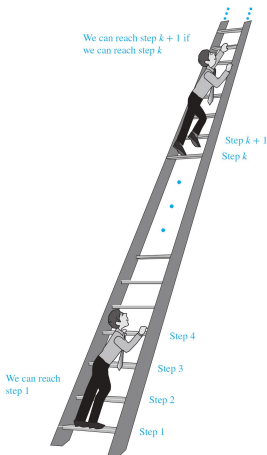
Proving $P(1)$ is called the *basis step* (usually, this step is easy).

Proving that $P(k) \implies P(k + 1)$ is called the *inductive step*.

In the inductive step, $P(k)$ is called the *inductive hypothesis*.

Mathematical induction

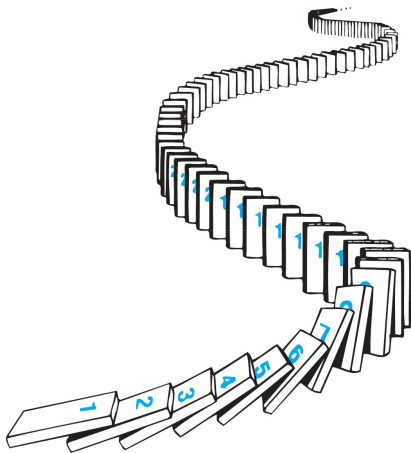
This may look like circular reasoning, but it's not.



We are simply saying that, if we start from step 1, and from any step k we can always reach step $k + 1$, then we can reach all steps.

Mathematical induction

The same happens with domino tiles.



If the first tile falls, and each tile causes the next tile to fall, then eventually all tiles will fall.

Exercise

Prove the following statements by mathematical induction.

- $1 + 3 + 5 + 7 + \cdots + (2n - 1) = n^2$
- $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$
- $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$
- $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$
- $n^2 + n$ is an even number.
- $n^3 + 2n$ is a multiple of 3.
- If S is a set of size $|S| = n$, then $|\mathcal{P}(S)| = 2^n$.
- Every $2^n \times 2^n$ checkerboard with one square removed can be tiled using right triominoes (a right triomino is a 2×2 tile with one 1×1 corner removed).